[METRIC]
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June 10, 1997
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

LINING BRAKE, AND LINING MATERIAL, BRAKE: AUTOMOTIVE (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers automotive brake lining sets and brake lining material used in commercial and military vehicles. Except for the following: heavy construction, heavy military, industrial aeronautical, and any other application where the friction material is compounded specifically for a special purpose vehicle.

2. CLASSIFICATION. Brake linings shall be classified as follows (see 7.2):

Type I - Brake lining sets, rigid (molded) materials.
Type II - Brake lining material flexible (woven).
Type III - Brake lining material flexible (molded).

NOTE: 1. A rigid lining is formed to fit a specific drum or disc diameter.
2. A woven lining is one which is made with woven fabric as a base.
3. A flexible lining is one that is adaptable to brake drums, the diameter of which are equal to or greater than 250 millimeters (mm) [10 inches (in.)].

Category A - Drum brakes.
Category B - Disc brakes.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 2530

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Style I - For bonding.
Style II - Drilled for rivets or bolts.

Class L - Up to 2727 kilograms (kg) [Up to 6000 pounds (lbs)].
Class M - 2728 to 4545 kg (6001 to 10 000 lbs).
Class H - Over 4545 kg (over 10 000 lbs).

3. SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein; materials shall be in accordance with the manufacturer's material specification. Asbestos material shall not be used. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 4.1).

3.1.1 Stability. Lining shall not vary more than as specified in Table I, subsequent to testing in accordance with SAE J160.

TABLE I. Allowable variation for lining thickness.

Lining size		Maximum variation in thickness	
mm	in.	mm	in.
Up to 10	Up to 0.38	0.3	0.010
10 - 19	0.38 - 0.75	0.4	0.015
Above 19	Above 0.75	0.5	0.020

3.1.1.1 Gogan hardness and specific gravity. Unless otherwise specified in the contract (see 7.2), Gogan (or other) hardness shall be certified by the contractor to meet the levels specified in SAE J379. Specific gravity shall be determined by using the procedure described in SAE J380.

3.1.2 Sets or roll(s). Set(s) shall consist of all components required for one axle, shall be preformed and prepared for attachment to specific brake shoes as required (see 7.2). Rolls shall consist of bulk material furnished in lengths, widths and thicknesses as specified (see 7.2). Sets shall be attached, or be prepared for attachment, in one of the following ways:

a. Drilled with rivet or bolt holes for attachment to specific brake shoes.
b. Undrilled, for bonding of lining to the backing shoes.
c. Molded or mechanically attached disc brake pads.

3.1.3 Bonding agent. The bonding agent shall be of the thermo-setting variety. Lining applied to the shoe or pad shall have a shear force exerted to the bond joint per SAE J840. Shear strength of the bonding agent shall be certified by the contractor to meet agreed upon levels. The agent, when cured, shall be unaffected by grease, oil, brake fluid, cleaning fluid, or storage of up to

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82 degrees Celsius (°C) [180 degrees Fahrenheit (°F)] for periods of up to 2 years. The contractor shall certify to these requirements.

3.1.4 Storage. Both the adhesive in its final cured state, and the lining, after being bonded to the shoe, shall show no undesirable change in properties after periods of up to two years of storage at temperatures ranging from minus (-) 54°C (-65°F) to plus (+) 82°C (180°F). Storage capability shall be certified by the contractor.

3.1.5 Mechanical strength. Lining shall have sufficient mechanical strength to permit secure attachment without damage during assembly or use.

3.2 Design and construction. Unless otherwise specified herein, the design and construction of the brake lining shall be in accordance with the manufacturer's design standards.

3.2.1 Brake lining tolerance for drum brakes.

3.2.1.1 Lining thickness. Except when variation in thickness is a function of design, surfaces of the drum brake lining shall be essentially parallel. Maximum variation in thickness across the width of the lining shall not be greater than ± 0.2 mm (± 0.008 in.)

3.2.1.2 Rivets and bolts. When riveted or bolted material is specified, rivets or bolts conforming to SAE J663 shall be provided in the manner and size required for the applicable lining.

3.2.1.3 Radius of curvature. Unless otherwise specified (see 7.2), inside radius of rigid molded lining shall be held to ± 0.64 mm (± 0.025 in.).

3.2.1.4 Lining position. All lining segments shall be properly identified as to position, such as primary, secondary, forward, reverse, inner or outer. Intermixing of primary and secondary linings shall not be permitted.

3.2.2 Brake lining tolerances for disc brakes. Unless otherwise specified (see 7.2), tolerances for disc brake type linings shall be as follows:

a. Flat within 0.15 mm (0.005 in.) on lining surface, and parallel within 0.25 mm (0.010 in.) for integrally molded assemblies.
b. Flat within 0.25 mm (0.010 in.) on both lining surfaces when supplied blank or drilled, and parallel within 0.51 mm (0.020 in.) when assembled on the steel backing.

3.3 Performance. Lining shall perform as specified under the conditions described in SAE J661.

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3.3.1 Coefficients of friction. Values for coefficients of friction shall not be less than as specified in Table II for the types, categories and classes shown. To meet minimum requirements, a brake lining shall have:

a. Minimal normal friction coefficient as specified in Column a, based on the average of five tests.
b. Minimal hot friction coefficient as specified in Column b, based on the average of five tests.
c. Minimal coefficient of friction as specified in Column c, based on each of five tests at the following points:
   (1) Between 93°C (200°F) and 260°C (500°F), inclusive, on the second fade run.
   (2) Between 148°C (300°F) and 93°C (200°F), inclusive on the second recovery run.

3.3.2 Wear. Wear shall not exceed the values specified in Table II for the types, categories and classes shown.

TABLE II. Coefficients of friction and wear tolerances.

Type	Category	Class	Friction coefficient			Maximum wear	
			a	b	c	mm	in.
I	A	L	0.25	0.20	0.18	0.75	0.030
		M	0.25	0.20	0.18	0.8	0.035
		H	0.25	0.20	0.18	1.0	0.040
I	B	L	0.28	0.23	0.21	0.75	0.030
		M	0.28	0.23	0.21	0.8	0.035
		H	0.28	0.23	0.21	1.0	0.040
II & III	-	-	0.30	0.20	0.15	2.0	0.075

3.4 Identification and marking. Identification and marking of the brake lining shall be permanent and legible and shall include as a minimum, the manufacturer's CAGE code and part or identification number (PIN). Lining furnished in rolls, shall be marked at intervals not to exceed 305 mm (12 in.) throughout the length of the roll. Month and year of manufacture, by numerical code or other means, shall also be marked on each item, or at each interval.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. QUALITY ASSURANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, examinations, and tests.

5.2 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents.

7.1.1 Government publications. Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Non-Government Publications. Copies of SAE J160 "Swell, Growth, and Dimensional stability of Brake Linings"; SAE J379 "Gogan Hardness of Brake Lining"; SAE J380 "Specific Gravity of Brake Lining"; SAE J661 "Brake Lining Quality Control Test Procedure"; SAE J663 "Rivets For Brake Linings and Bolts for Brake Block"; and SAE J840 "Test Procedures for Brake Shoe and Lining Adhesives and Bonds" are available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Type, category, style and class of lining, (PIN) (see 2).
d. Quality control level of Gogan (or other) hardness and specific gravity (see 3.1.1.1).
e. Sets or rolls are to be furnished (see 3.1.2).
f. If inside radius is other than $\pm$ 0.64 mm ($\pm$ 0.025 in) (see 3.2.1.3).

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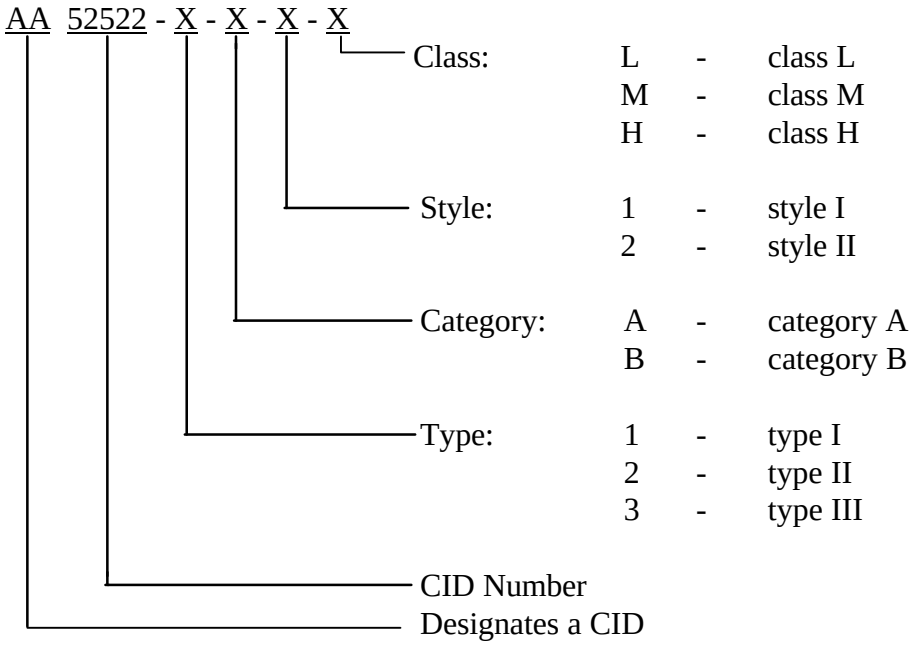
g. Disc brake lining tolerances (see 3.2.2).
h. Selection of applicable level and packaging requirements (see 6).

7.3 Part or identification number (PIN). The PIN to be used for brake linings acquired to this CID is as follows:



7.4 Cross reference data. Lining material conforming to this CID are interchangeable/ substitutable with lining materials conforming to HH-L-361G dated August 1, 1985.

MILITARY INTERESTS:

Custodians:
Army - AT
Air Force - 99

Review Activity:
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY
GSA - FSS

Preparing Activity:
Army - AT

(Project 2530-0395)

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